Tyler Bissonnette

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Electrical and Computer Engineering senior focused on analog and mixed-signal hardware design, PCB prototyping, and circuit debugging. Combines research in 4-layer PCB layout, LTspice circuit simulation, and bench characterization with an industry background in cleanroom semiconductor manufacturing.

EDUCATION

University at Albany, State University of New York (SUNY) — Albany, NY
Bachelor of Science in Electrical & Computer Engineering | Cumulative GPA: **3.4 / 4.0** — **Expected: Dec 2026**

Hudson Valley Community College (HVCC) — Troy, NY
Associate in Science (A.S.), Engineering Science — **2022 – 2024**

TECHNICAL SKILLS

Hardware & Prototyping: Analog Circuit Design, KiCad (2- and 4-Layer PCB Routing), Surface-Mount Soldering (down to 0402), Through-Hole Soldering, Stencil Printing & Reflow, Rapid Prototyping, 3D Printing, Hardware-Software Integration

Test & Measurement: Oscilloscopes, Differential Probes, Function Generators, Benchtop Power Supplies, Digital Multimeters (DMM), Circuit Debugging, Cleanroom ESD Protocols

Software & Simulation: SPICE / LTspice, C, C++, Python (NumPy, sounddevice), MATLAB, R, Autodesk Fusion (basic)

TECHNICAL & ENGINEERING PROJECTS

High-Voltage Solid-State Circuit Breaker & Gate Drive | *Undergraduate Research Assistant (SURP)* — Albany, NY • **June 2025 – Present**
Paid summer 2025 SURP position; continued for credit in spring 2026 and non-credit since.

- **Modeling & Simulation:** Derived RLC equations and swept L and C values in Python to find zero-current-crossing conditions for fast fault isolation (modeled for a 100 A fault current); verified the timing in LTspice.
- **PCB Design:** Designed 2- and 4-layer prototype PCBs in KiCad for the gate drivers and the circuit breaker power board (designed for 400–500 V); completed seven gate-drive boards.
- **Bench Verification:** Verified gate-driver outputs with a function generator and oscilloscope, then confirmed switching and current flow on the power board with differential probes (5 V source into a 1.6 Ω load, 3 A). Full-voltage testing is pending.
- **PCB Debugging:** Diagnosed and corrected design and assembly faults (wrong resistor values, an inverted transistor footprint, a capacitor footprint error) using DMM tracing and magnet-wire rework.

Continuous Sound Level Meter (SLM) | *Capstone Design Project* — Albany, NY • **2026 – Present**

- **Acoustic Instrumentation:** Designing a continuous sound level meter toward ANSI S1.4 and S1.11 Class-2 specifications, translating the standards' requirements into circuit and filtering constraints.
- **Sensor Front-End & DSP:** Interfaced an analog microphone and precision ADC with a Raspberry Pi 4B; implemented Python DSP filters (sounddevice) to compute A- and C-weighted sound levels (dB SPL), with the implementation still being refined.

Eurorack Clones & SMD Assembly | *Independent Engineering Project* — Albany, NY • **2023 – Early 2026**

- **SMD Assembly & Debug:** Assembled three Eurorack modules (Veils, Blades, Marbles) from purchased Mutable Instruments clone PCBs using stencil-printed solder paste, hand-placed 0402 SMD parts, hot-plate reflow, and iron/hot-air rework; hand-soldered all through-hole parts. Debugging with a bench oscilloscope: Veils fault traced to reversed diodes; Blades and Marbles unresolved when testing stopped in early 2026.
- **Electro-Mechanical Tooling:** Built a cordless vacuum solder-paste stencil fixture (impeller, drone motor, flight controller, LiPo power) from open-source CAD models, an online video guide, a BOM and poorly documented wiring, with a custom PCB holder for each board size; calibrated an Ender 3 for dimensional accuracy and iterated the ~30-hour print.

PROFESSIONAL EXPERIENCE

GlobalFoundries | *CMP Operator & Technical Trainer* — Essex Junction, VT • **June 2020 – Sept 2021**

- **Cleanroom Operations:** Operated CMP (chemical-mechanical planarization) equipment in a semiconductor cleanroom in full gowning, following SOPs and ESD and contamination-control protocols.
- **Wafer Handling & Quals:** Ran daily qualifications on multiple tools; handled wafers manually with wands, visually inspected them, and hand-carried wafer lots.
- **Technical Leadership:** Promoted to New-Hire Technical Trainer to onboard, instruct, and qualify incoming operators on tooling safety, SOPs, and quality metrics.